another one is plain hole. The bolt is screwed into a threaded hole without the nut, while it passes freely through a plain hole on the upper part.



Tap Bolt

Stud or Stud Bolt: Bolts that do not have an head is called as stud bolt. The center of the bolt may have a collar or square section. The stud is threaded at both ends. One end of the stud is screwed into a tapped hole of the parts to be fastened, while the other end receives a nut on it. It is used to cover the engine and pump cylinders, valves etc.,



Stud Bolt

NUTS

A element used with a bolt or a stud to join two or more parts together temporarily is known as a nut. Nut is a square or hexagonal shape having a threaded through hole. In order to ensure proper joint of the connection, a cotter pin is inserted into a hole made across the bolt diametrically.



Nuts

WASHER

A circular disc of a metal, having a hole in centre for inserting a bolt is known as a washer. It is placed below a nut to provide a perfect seating for the nut. The washer is generally specified by its hole diameter.



Washer

Uses of Washer

1. To provide perfect seating
2. The pressure applied on the nut is limited
3. If the hole is bigger than the head of the bolt, the washer is used

The washers are classified into two types as follows

1. Plain Washer
2. Spring Washer



Plain washer and Spring Washer

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